

# ABHAY GUPTA

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## EDUCATION

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### M. S. Ramaiah Institute of Technology

Bachelor's of Engineering in Electronics and Communication – CGPA: 8.24

2023 – 2027

Bengaluru, India

### Nalanda Academy

Senior Secondary Education (XII) – Percentage: 91.6%

2021 – 2022

Kota, Rajasthan

### JPV DAV Public School

Secondary Education (X) – Percentage: 90%

2019 – 2020

Katni, Madhya Pradesh

## PROJECTS

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### HRV-Based Biofeedback System | *ESP32, MAX30102 Sensor, C/C++*

- Architected a real-time HRV monitoring system using **ESP32** and the **MAX30102 sensor** for continuous physiological signal acquisition.
- Incorporated **RMSSD-based heart rate variability analysis** with guided breathing feedback, improving stress regulation by approximately 25%.
- Developed C/C++ noise-filtering and sampling algorithms that stabilized heart-rate measurements by filtering motion artifacts across **500+ sensor samples**.
- Integrated **embedded signal processing** with live tracking to deliver persistent **physiological biofeedback**, facilitating enhanced **autonomic nervous system** modulation.

### [RepoFlow](#) | *JavaScript, GitHub OAuth, GitHub REST API*

- Constructed a web application integrating **GitHub OAuth authentication** with the platform's **REST API** for secure repository management.
- Empowered users to browse 50+ repositories instantly through an intuitive layout, eliminating repetitive manual repository searches.
- Managed the deployment on **Vercel** with a responsive design, ensuring optimized performance and seamless cross-platform accessibility.

### AI Resume Screening and Candidate Ranking System | *Python, Groq API, Llama 3.3-70B, Pydantic, PyPDF, python-docx*

- Crafted an **LLM-powered Applicant Tracking System (ATS)** that parsed PDF and DOCX CVs into 10+ structured candidate attributes using schema-guided JSON generation.
- Fashioned an **AI matching engine** that compared resumes with job descriptions, generated 0–100 compatibility scores, identified missing skills, and reduced manual resume screening effort by approximately 60%.
- Leveraged Pydantic schemas to validate **10+ data points** from PDF and DOCX documents, ensuring consistent LLM output across document formats.
- Optimized hiring workflows by implementing **batch processing** and **automated candidate ranking**, providing actionable recruitment summaries to expedite talent acquisition.

## TECHNICAL SKILLS

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**Languages:** C, C++, Python, JavaScript, HTML, CSS

**Libraries/APIs:** Groq API, GitHub REST API, Pydantic, NumPy, Matplotlib

**Tools:** Git, GitHub, Vercel, Overleaf, Microsoft Excel

**Core Concepts:** Data Structures, OOPS Concepts, API Integration, Embedded Systems, Problem Solving

## CERTIFICATIONS AND TEAMWORK

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- IISc Workshop on PCB Design using Eagle
- The Bits and Bytes of Computer Networking (Google) - Coursera
- Served as an IEEE Executive Committee (ExeCom) member during my 4th semester, coordinating 10+ technical events.